Wireshark Network Analysis

Project Overview

This project demonstrates network packet analysis using Wireshark. Different types of network packets such as HTTP, DNS, TCP and ICMP were analysed to understand network communication.

Tools Used

Wireshark Kali Linux

Tasks Performed

HTTP Packet Analysis

- Captured HTTP packets and extracted User-Agent information.

DNS Traffic Analysis

- Filtered DNS packets and identified DNS transaction IDs.

TCP Three Way Handshake

- Analysed SYN, SYN-ACK and ACK packets to understand TCP connection process.

ICMP Packet Analysis

- Captured ICMP packets and identified source IP and MAC addresses.

Skills Gained

Network Packet Analysis

TCP/IP Protocol Understanding

Network Security Investigation

- What is the User-Agent part of HTTP GET request sent to IP address 10.10.10.12. (Hint : apply display filter to list http get requests.)

The **User-Agent** in an HTTP GET request is a string sent in the request headers that provides information about the client making the request (Operating system or Device).

To view the User-Agent part of an HTTP GET request sent to IP 10.10.10.12, we can use a **packet capture tool** like **Wireshark** and apply a **display filter** to view only HTTP GET requests.

Steps in Wireshark:

1. **Apply a display filter** to only show HTTP GET requests to the IP address 10.10.10.12. we can use the following filter:

http.request.method==GET

No.	Time	Source	Destination	Protocol	Length	Info
527	112.951526	192.168.0.143	167.71.226.188	HTTP	386	GET / HTTP/1.1
2901	125.640309	192.168.0.143	104.21.65.93	HTTP	454	GET / HTTP/1.1
4459	165.026023	192.168.0.219	192.168.0.143	HTTP	478	GET /guess_password.png HTTP/1.1
4497	175.291309	192.168.0.219	192.168.0.143	HTTP	514	GET /guess_password.png HTTP/1.1
4909	14758.26960…	10.10.10.11	10.10.10.12	HTTP	225	GET /ellen_key.tar.gz HTTP/1.1

2. **Inspect the GET request**:
   - Select one of the HTTP GET request packets in the filtered results.
   - Look for the User-Agent field under the **HTTP header** section.

The User-Agent will be listed as part of the HTTP headers.

Wireshark Output:

```
▾ Hypertext Transfer Protocol
  ▸ GET /ellen_key.tar.gz HTTP/1.1\r\n
    User-Agent: Wget/1.20.3 (linux-gnu)\r\n
    Accept: */*\r\n
    Accept-Encoding: identity\r\n
    Host: 10.10.10.12:9000\r\n
    Connection: Keep-Alive\r\n
    \r\n
    [Full request URI: http://10.10.10.12:9000/ellen_key.tar.gz]
    [HTTP request 1/1]
    [Response in frame: 4913]
```

2. Found any image in the Wireshark capture? What is the extension? (Hint : how to export images?)

In Wireshark, it is possible to extract images that were transferred over network protocols such as HTTP. If images were transmitted during the capture session, we can export them and identify the file extensions.

Steps to Find and Export Images:

1. **Open Wireshark Capture**: Load the .pcap file start capturing traffic on the relevant network interface.
2. **Filter HTTP Objects**: To narrow down the search for images, we can apply the following display filter in Wireshark:

http

No.	Time	Source	Destination	Protocol	Length	Info
96	17.655845	192.168.0.143	123.176.33.17	OCSP	479	Request
98	17.658168	123.176.33.17	192.168.0.143	OCSP	954	Response
527	112.951526	192.168.0.143	167.71.226.188	HTTP	386	GET / HTTP/1.1
529	112.963492	167.71.226.188	192.168.0.143	HTTP	608	HTTP/1.1 302 Found (text/html)
564	113.446107	192.168.0.143	192.124.249.23	OCSP	471	Request
567	113.453100	192.168.0.143	192.124.249.23	OCSP	471	Request
578	113.972831	192.124.249.23	192.168.0.143	OCSP	2353	Response
583	113.980542	192.124.249.23	192.168.0.143	OCSP	2353	Response
748	118.094637	192.168.0.143	142.250.196.67	OCSP	483	Request
758	118.145167	142.250.196.67	192.168.0.143	OCSP	768	Response
1674	119.423416	192.168.0.143	142.250.196.67	OCSP	482	Request
1678	119.470177	142.250.196.67	192.168.0.143	OCSP	767	Response

3. **Export HTTP Objects**:
   - Go to **File** > **Export Objects** > **HTTP**.

4. **Export the Image**:
    - Select the image file from the list and click **Save As** to export it to our local machine.

HSCS01{bfd3f4c834aac369f9cc4b458f24feb0}

3. What is the Transaction ID of DNS standard query for hacker (A record)? (Hint : apply protocol based display filter.)

To find the **Transaction ID** of a DNS standard query for the domain "hacker" (A record) in Wireshark, follow these steps:

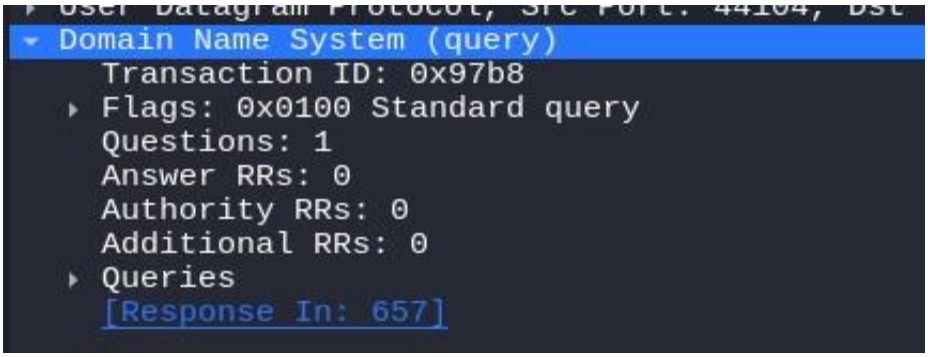
Steps:

1. **Open Wireshark**: Load the packet capture (.pcap) file or start capturing traffic on the network interface.
2. **Apply a Display Filter**: Use the following filter to show only DNS queries for "hacker":

dns.qry.name == hacker

No.	Time	Source	Destination	Protocol	Length	Info
650	117.661647	192.168.0.143	192.168.0.1	DNS	66	Standard query 0x1693 A hacker
651	117.661671	192.168.0.143	192.168.0.1	DNS	66	Standard query 0x0491 AAAA hacker
652	117.664147	192.168.0.1	192.168.0.143	DNS	141	Standard query response 0x1693 No such name A hacker SOA a.root-servers.net
653	117.664148	192.168.0.1	192.168.0.143	DNS	141	Standard query response 0x0491 No such name AAAA hacker SOA a.root-servers.net
654	117.664413	192.168.0.143	192.168.0.1	DNS	66	Standard query 0x97b8 A hacker
655	117.664431	192.168.0.143	192.168.0.1	DNS	66	Standard query 0x48b9 AAAA hacker
656	117.665847	192.168.0.1	192.168.0.143	DNS	141	Standard query response 0x48b9 No such name AAAA hacker SOA a.root-servers.net
657	117.666692	192.168.0.1	192.168.0.143	DNS	141	Standard query response 0x97b8 No such name A hacker SOA a.root-servers.net

3. **Locate the Transaction ID**:
    - In the filtered list, look for the DNS query packet.
    - Select the packet and in the **Packet Details** window, expand the **Domain Name System (DNS)** section.
    - The **Transaction ID** will be listed at the top of this section as a hexadecimal number.



4. What is the User-Agent part of HTTP HEAD request with source and destination address as loopback address. (Hint : apply display filter to list http packets with loopback address.)

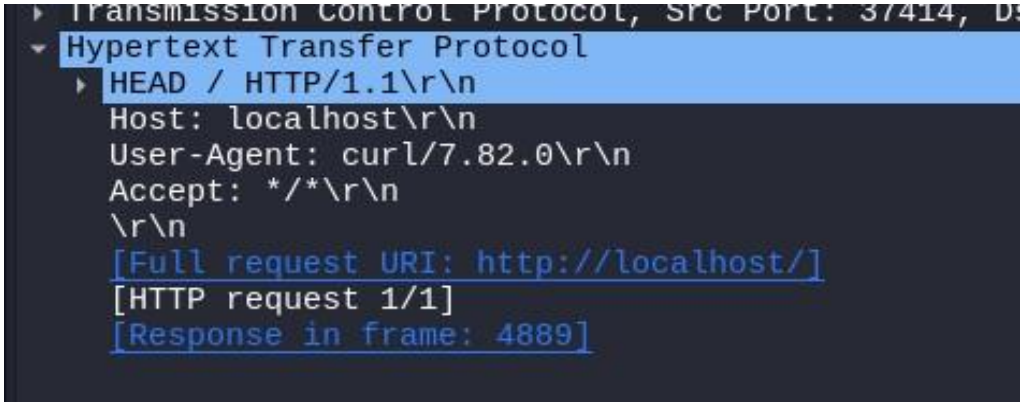
The **User-Agent** part of an HTTP HEAD request where both the source and destination addresses are the loopback address (127.0.0.1), we can apply a specific display filter in Wireshark to isolate such packets.

Steps:

1. **Open Wireshark**: Load the capture file or start a new capture.
2. **Apply a Display Filter**: Use the following display filter to find HTTP HEAD requests with the loopback (127.0.0.1) as both the source and destination address:

This filter will display only HTTP HEAD requests where both the source and destination IP addresses are the loopback (127.0.0.1).

3. **Locate the User-Agent**:
    - In the filtered results, select the HTTP HEAD request packet.
    - Expand the **Hypertext Transfer Protocol** section in the **Packet Details** pane.
    - Scroll down to the **User-Agent** field under the HTTP headers.



5. What is the IP address and MAC address of the machine which initiated the ICMP request? (Hint : use wireshark display filter.)

The **IP address** and **MAC address** of the machine that initiated an ICMP request, we can apply a specific display filter in Wireshark and examine the relevant packet details.

Steps to Find IP and MAC Address of ICMP Request Initiator:

1. **Open Wireshark**: Load the capture file or start a new capture session.

2. **Apply a Display Filter**: Use the following display filter to show only ICMP request packets.

icmp

No.	Time	Source	Destination	Protocol	Length	Info
4374	157.798255	192.168.0.143	192.168.0.1	ICMP	120	Destination unreachable (Port unreachable)
4376	157.810172	192.168.0.143	192.168.0.1	ICMP	120	Destination unreachable (Port unreachable)
4634	187.855765	192.168.0.143	192.168.0.177	ICMP	98	Echo (ping) request id=0x28a8, seq=1/256, ttl=64 (no response found!)
4647	188.860477	192.168.0.143	192.168.0.177	ICMP	98	Echo (ping) request id=0x28a8, seq=2/512, ttl=64 (no response found!)
4648	189.884997	192.168.0.143	192.168.0.177	ICMP	98	Echo (ping) request id=0x28a8, seq=3/768, ttl=64 (no response found!)
4652	190.908041	192.168.0.143	192.168.0.177	ICMP	98	Echo (ping) request id=0x28a8, seq=4/1024, ttl=64 (no response found!)
4739	219.854264	192.168.0.143	192.168.0.219	ICMP	98	Echo (ping) request id=0xdf34, seq=1/256, ttl=64 (no response found!)
4747	220.872040	192.168.0.143	192.168.0.219	ICMP	98	Echo (ping) request id=0xdf34, seq=2/512, ttl=64 (no response found!)
4758	221.883973	192.168.0.143	192.168.0.219	ICMP	98	Echo (ping) request id=0xdf34, seq=3/768, ttl=64 (no response found!)
4782	228.555405	192.168.0.143	192.168.0.1	ICMP	120	Destination unreachable (Port unreachable)
4784	228.567170	192.168.0.143	192.168.0.1	ICMP	120	Destination unreachable (Port unreachable)

3. **Inspect the Packet**:
   - Select one of the filtered packets.
   - In the **Packet Details** pane, expand the **Internet Protocol Version 4 (IPv4)** section to see the source IP address (which is the address of the machine that initiated the ICMP request).
   - we can also expand the **Ethernet II** section to find the source **MAC address**.

```
▸ Frame 4758: 98 bytes on wire (784 bits), 98 bytes captured (784 bits) on interface unknown, id 0
▸ Ethernet II, Src: PCSSystemtec_86:1b:0b (08:00:27:86:1b:0b), Dst: PCSSystemtec_a2:ab:4b (08:00:27:a2:ab:4b)
▸ Internet Protocol Version 4, Src: 192.168.0.143, Dst: 192.168.0.219
```

6. What is the Sequence Number (raw) in the last packet part of TCP three way-handshake established between 192.168.0.143 and 34.98.75.36 (Hint : expand packet details pane.)

To find the Sequence Number (in raw format) in the last packet of a TCP three-way handshake between the IP addresses 192.168.0.143 and 34.98.75.36 using Wireshark, follow these steps:

Steps to Identify the Sequence Number:

1. **Open the Packet Capture File** in Wireshark:
    - Open the capture file (.pcap or .pcapng) that contains the traffic in Wireshark.
2. **Apply a Display Filter** to isolate the handshake:
    - Use the following display filter to show only TCP packets between 192.168.0.143 and 34.98.75.36:

ip.src == 192.168.0.143 && ip.dst == 34.98.75.36 && tcp

No.	Time	Source	Destination	Protocol	Length	Info
80	17.611442	192.168.0.143	34.98.75.36	TCP	74	44316 → 443 [SYN] Seq=0 Win=64240 Len=0 MSS=1460 SACK_PERM TSval=4187266083 TSecr=0 WS=128
82	17.628869	192.168.0.143	34.98.75.36	TCP	66	44316 → 443 [ACK] Seq=1 Ack=1 Win=64256 Len=0 TSval=4187266100 TSecr=3306572767
83	17.630915	192.168.0.143	34.98.75.36	TLSv1.3	583	Client Hello (SNI=classify-client.services.mozilla.com)
87	17.648989	192.168.0.143	34.98.75.36	TCP	66	44316 → 443 [ACK] Seq=518 Ack=2837 Win=63488 Len=0 TSval=4187266120 TSecr=3306572787
88	17.649012	192.168.0.143	34.98.75.36	TCP	66	44316 → 443 [ACK] Seq=518 Ack=4540 Win=61824 Len=0 TSval=4187266120 TSecr=3306572787
100	17.659850	192.168.0.143	34.98.75.36	TLSv1.3	130	Change Cipher Spec, Application Data
101	17.660172	192.168.0.143	34.98.75.36	TLSv1.3	236	Application Data
102	17.660192	192.168.0.143	34.98.75.36	TLSv1.3	321	Application Data
106	17.674484	192.168.0.143	34.98.75.36	TCP	66	44316 → 443 [ACK] Seq=1007 Ack=5120 Win=64128 Len=0 TSval=4187266146 TSecr=3306572816
107	17.674508	192.168.0.143	34.98.75.36	TCP	66	44316 → 443 [ACK] Seq=1007 Ack=5151 Win=64128 Len=0 TSval=4187266146 TSecr=3306572816
108	17.674866	192.168.0.143	34.98.75.36	TLSv1.3	97	Application Data
113	17.860899	192.168.0.143	34.98.75.36	TCP	66	44316 → 443 [ACK] Seq=1038 Ack=5304 Win=64128 Len=0 TSval=4187266332 TSecr=3306573001
114	17.860917	192.168.0.143	34.98.75.36	TCP	66	44316 → 443 [ACK] Seq=1038 Ack=5430 Win=64128 Len=0 TSval=4187266332 TSecr=3306573001
115	17.860924	192.168.0.143	34.98.75.36	TCP	66	44316 → 443 [ACK] Seq=1038 Ack=5469 Win=64128 Len=0 TSval=4187266332 TSecr=3306573002
116	17.862417	192.168.0.143	34.98.75.36	TLSv1.3	105	Application Data
416	76.634956	192.168.0.143	34.98.75.36	TLSv1.3	105	Application Data
421	76.715205	192.168.0.143	34.98.75.36	TCP	66	44316 → 443 [ACK] Seq=1116 Ack=5508 Win=64128 Len=0 TSval=4187325186 TSecr=3306631792
3452	135.083204	192.168.0.143	34.98.75.36	TLSv1.3	105	Application Data
3456	135.100735	192.168.0.143	34.98.75.36	TCP	66	44316 → 443 [ACK] Seq=1155 Ack=5547 Win=64128 Len=0 TSval=4187383572 TSecr=3306690240
4636	188.682166	192.168.0.143	34.98.75.36	TLSv1.3	105	Application Data
4637	188.682377	192.168.0.143	34.98.75.36	TLSv1.3	90	Application Data
4638	188.682396	192.168.0.143	34.98.75.36	TCP	66	44316 → 443 [FIN, ACK] Seq=1218 Ack=5547 Win=64128 Len=0 TSval=4187437154 TSecr=3306690240
4646	188.700032	192.168.0.143	34.98.75.36	TCP	66	44316 → 443 [ACK] Seq=1219 Ack=5548 Win=64128 Len=0 TSval=4187437171 TSecr=3306743840

3. **Locate the Three-Way Handshake Packets**:
    - In a TCP three-way handshake, there are three main packets:
        1. **SYN**: Sent from the client to initiate the connection.
        2. **SYN-ACK**: Sent by the server in response to the SYN, acknowledging it.
        3. **ACK**: Sent by the client to acknowledge the SYN-ACK from the server.
        4. **Find the Sequence Number in the Last ACK Packet**:
    - Select the final **ACK** packet in the three-way handshake.

- Expand the **Transmission Control Protocol** section in the packet details pane.
- Look for the **Sequence Number** field, which is typically labeled as:
  - **[Sequence number (raw): X]**

5. **The Raw Sequence Number**:
   - The raw sequence number in the final ACK packet of the threeway handshake. This is the required value for the task.

Output:-

```
▸ Internet Protocol Version 4, Src: 192.168.0.143, Dst: 34.98.75.36
▾ Transmission Control Protocol, Src Port: 44316, Dst Port: 443, Seq: 1219, Ack: 5548, Len: 0
    Source Port: 44316
    Destination Port: 443
    [Stream index: 5]
  ▸ [Conversation completeness: Complete, WITH_DATA (31)]
    [TCP Segment Len: 0]
    Sequence Number: 1219    (relative sequence number)
    Sequence Number (raw): 969756235
    [Next Sequence Number: 1219    (relative sequence number)]
    Acknowledgment Number: 5548    (relative ack number)
    Acknowledgment number (raw): 3052908760
    1000 .... = Header Length: 32 bytes (8)
  ▸ Flags: 0x010 (ACK)
    Window: 501
    [Calculated window size: 64128]
    [Window size scaling factor: 128]
    Checksum: 0xc905 [unverified]
    [Checksum Status: Unverified]
    Urgent Pointer: 0
  ▸ Options: (12 bytes), No-Operation (NOP), No-Operation (NOP), Timestamps
  ▸ [Timestamps]
  ▸ [SEQ/ACK analysis]
```

7. What is the Sequence Number (raw) in the second packet part of TCP three-way-handshake established between 192.168.0.143 and hackerschool.in (Hint : apply display filter and select tcp stream.)

To find the Sequence Number (in raw format) in the second packet of a TCP three-way handshake between the IP address 192.168.0.143 and the domain hackerschool.in, we can use Wireshark by following these steps:

Steps to Identify the Sequence Number:

1. **Open Wireshark** and Load the Capture File:
   - Open our network capture file (.pcap or .pcapng) in Wireshark.

2. **Apply a Display Filter to Focus on the TCP Handshake**:

   - First, resolve the IP address of hackerschool.in if it's not already shown in the capture (Wireshark may automatically resolve it). Alternatively, we can look up the IP using the command line:

ping hackerschool.in

```
┌──(kali㉿kali)-[~]
└─$ ping hackerschool.in
PING hackerschool.in (104.21.65.93) 56(84) bytes of data.
64 bytes from 104.21.65.93: icmp_seq=1 ttl=59 time=71.7 ms
64 bytes from 104.21.65.93: icmp_seq=2 ttl=59 time=97.3 ms
64 bytes from 104.21.65.93: icmp_seq=3 ttl=59 time=127 ms
64 bytes from 104.21.65.93: icmp_seq=4 ttl=59 time=53.6 ms
64 bytes from 104.21.65.93: icmp_seq=5 ttl=59 time=5.32 ms
^C
--- hackerschool.in ping statistics ---
5 packets transmitted, 5 received, 0% packet loss, time 4011ms
rtt min/avg/max/mdev = 5.315/70.970/126.901/41.066 ms
```

   - Suppose hackerschool.in resolves to 104.21.65.93 Apply this display filter to isolate the TCP packets exchanged between 192.168.0.143 and hackerschool.in:

ip.src == 192.168.0.143 && ip.dst == 104.21.65.93 && tcp

No.	Time	Source	Destination	Protocol	Length	Info
2898	125.621826	192.168.0.143	104.21.65.93	TCP	74	49946 → 80 [SYN] Seq=0 Win=64240 Len=0 MSS=1460 SACK_PERM TSval=935558972
2900	125.623459	192.168.0.143	104.21.65.93	TCP	54	49946 → 80 [ACK] Seq=1 Ack=1 Win=64256 Len=0
2901	125.640309	192.168.0.143	104.21.65.93	HTTP	454	GET / HTTP/1.1
2902	125.640709	192.168.0.143	104.21.65.93	TCP	54	49946 → 80 [FIN, ACK] Seq=401 Ack=1 Win=64256 Len=0
2905	125.642958	192.168.0.143	104.21.65.93	TCP	54	49946 → 80 [ACK] Seq=402 Ack=2 Win=64256 Len=0

3. **Identify the Three-Way Handshake Packets**:

   - In the display pane, look for the three-way handshake packets in the following sequence:

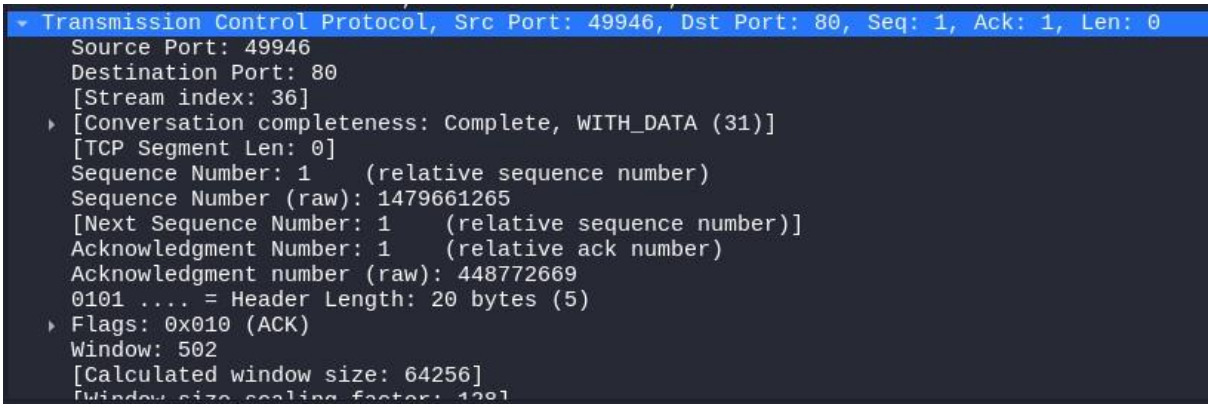
     1. **SYN**: Sent from the client (192.168.0.143) to hackerschool.in to initiate the connection.

     2. **SYN-ACK**: Sent by hackerschool.in to acknowledge the SYN.

3. **ACK**: Sent by the client to acknowledge the SYN-ACK, completing the handshake.

4. **Find the Sequence Number in the SYN-ACK Packet**:
   - Select the **SYN-ACK** packet (second packet in the three-way handshake) in the packet list.
   - Look for the **Sequence Number** field, which will display the raw sequence number as:
     - **[Sequence number (raw): X]** ○ This raw

sequence number in the SYN-ACK packet



```
▾ Transmission Control Protocol, Src Port: 49946, Dst Port: 80, Seq: 1, Ack: 1, Len: 0
    Source Port: 49946
    Destination Port: 80
    [Stream index: 36]
  ▸ [Conversation completeness: Complete, WITH_DATA (31)]
    [TCP Segment Len: 0]
    Sequence Number: 1    (relative sequence number)
    Sequence Number (raw): 1479661265
    [Next Sequence Number: 1    (relative sequence number)]
    Acknowledgment Number: 1    (relative ack number)
    Acknowledgment number (raw): 448772669
    0101 .... = Header Length: 20 bytes (5)
  ▸ Flags: 0x010 (ACK)
    Window: 502
    [Calculated window size: 64256]
```